medium-term, and long-term.[32] Most fund managers, especially active managers seeking alpha, have short time horizons and work in a crowded space. At the other end, asset owners represented by boards, a Ministry of Finance, trustees, or investment committees usually set long-term asset allocation weights. For Norway and CPPIB, this is the long-term equity-bond allocation set at the highest levels, which are Parliament and the board, respectively.

Dynamic factor strategies occupy the middle ground as their verification horizons are in the order of two to five years. Value strategies, for instance, underperformed for up to five years during the late 1990s in the midst of the Internet mania. Many short-term asset managers harvest factor risk premiums (which sometimes masquerade as alpha). If the asset owner does not know the factor exposures, then she can be surprised and angry—as the Norwegian public was with NBIM—when the factors perform badly. *Factor benchmarks do not penalize a manager for losses that are due to poor factor performance.*

Asset owners also benefit from explicit factor benchmarks if they choose to set portions of their portfolio to factor exposures. The ATP Group, Denmark's largest pension fund, is responsible for financing the retirement of most working Danes. ATP divides its financial assets into a hedging portfolio, which is managed to match the funds' liabilities, and an investment portfolio, whose aim "is to generate an absolute return that is sufficient to preserve the long-term purchasing power of pensions."[33] The investment portfolio is itself divided into risk groups: interest rates, credit, equities, inflation-linked assets, and commodities. ATP rebalances these factors to maintain approximate risk allocations (it practices a form of *risk parity*; see chapter 3). It also practices factor timing. The factor benchmarks allow ATP to invest counter-cyclically, so that factor exposures are increased as a factor strategy looks attractive.

Problems arise if neither the asset owner nor the fund manager takes responsibility for dynamic factors in the medium-term gray area. Many asset owners, for example, just gave up on volatility strategies after large losses in 2008, which was right at the time a counter-cyclical investor would have been increasing exposure (see chapters 2 and 4). Bringing a volatility risk factor, like all dynamic factors, into the fund's overall benchmark would enhance asset owners' ability to commit to the factor strategy over the long haul.

4.6. SUMMARY

Factor investing leads to optimal nonmarket capitalization indexes. Investors ought to be able to invest in cheap factor portfolios and customize their factor exposures to investor-specific circumstances and characteristics. Innovation in the asset management industry is taking us closer to this optimal case.

[32] Figure 14.7 is adapted from Knut Kjær.

[33] From the 2011 annual report of the ATP Group.